

Mann §4.2–4.3 — Noether’s Theorem & Spacetime Currents

A Plain–Language Walkthrough

Reading companion — Mann, *An Introduction to Particle Physics and the Standard Model*, Ch. 4

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Where we are. In §4.1 we learned the *action principle*: the true path of a system extremizes the action $S[q] = \int dt L[q, \dot{q}]$, which gives the Euler–Lagrange equations and defines the canonical momentum $p_i = \partial L / \partial \dot{q}_i$. We also saw that if a transformation leaves S unchanged (*invariance*), the equations of motion keep their form (*covariance*). Now §4.2 turns that into the single most useful theorem in the chapter, and §4.3 reads off the three classic conservation laws from it.

1. §4.2 Noether’s Theorem — the one-sentence version

The whole idea

Every continuous symmetry of the action comes with a quantity that does not change in time. Symmetry in $\Rightarrow$ conservation law out. (Emmy Noether, 1917.)

Why should that be true? The proof is short once you spot the trick.

1.1 The trick: allow a “sloppy” variation

Back in §4.1 we varied the path but imposed two tidy conditions: the equations of motion held, *and* the endpoints were pinned ($\delta q = 0$ at the ends). Noether’s proof does the opposite — it allows a *fully arbitrary* variation, with *neither* condition imposed. Carrying the algebra of §4.1 one step further (an integration by parts, but now *keeping* the boundary term) gives the master identity

$$\delta_{q_i} L = \underbrace{\left[\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right]}_{\text{Euler–Lagrange bracket}} \delta q_i + \frac{d}{dt} [p_i \delta q_i]. \quad (1)$$

How to read eq. (1)

The change in the Lagrangian splits into two pieces:

- the **Euler–Lagrange bracket** — this is *zero whenever the real equations of motion hold*;
- a **total time derivative** $\frac{d}{dt} [p_i \delta q_i]$ — a “boundary” piece that is usually *not* a pure d/dt of anything nice.

1.2 Now feed in a symmetry

Describe a small symmetry by parameters $\alpha = \{\alpha_1, \dots, \alpha_m\}$, with $\alpha = 0$ meaning “do nothing” (the identity). The induced change in the coordinates is

$$\delta_{\alpha_j} q_i = \left. \frac{\partial q'_i}{\partial \alpha_j} \right|_{\alpha=0} \delta \alpha_j. \quad (2)$$

Here is the pivotal fact that makes a symmetry *special*:

Symmetry $\Leftrightarrow \delta L$ is a total time derivative

If the transformation is a genuine symmetry, then S is unchanged, so $\delta_\alpha S = \int dt \delta_\alpha L = 0$. The most general way an integral can vanish for arbitrary limits is for the integrand itself to be a total derivative:

$$\delta_{\alpha_k} L = \frac{d}{dt} G \left(q, \frac{\partial q'}{\partial \alpha} \right), \quad (3)$$

where G is some function (vanishing at the endpoints) that you must work out **case by case**. For a symmetry $\delta_\alpha L$ is always a total d/dt ; for a random variation it is not. That single difference is the engine of the whole theorem.

1.3 Build the conserved current

Combine the two total-derivative facts. Define the **Noether current**

$$J_k = p_i \frac{\partial q'_i}{\partial \alpha_k} - G \quad (\text{one } J_k \text{ per symmetry parameter } \alpha_k). \quad (4)$$

Differentiate it in time. Using eq. (1) and imposing the Euler–Lagrange equations (now we *do* sit on a real solution), the Euler–Lagrange bracket dies, and the leftover is exactly $\frac{d}{dt} G$, which cancels the $-G$ term:

$$\frac{dJ_k}{dt} = \frac{d}{dt} \left(p_i \frac{\partial q'_i}{\partial \alpha_k} \right) - \frac{dG}{dt} = \underbrace{\left[\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right]}_{=0 \text{ (EOM)}} \frac{\partial q'_i}{\partial \alpha_k} + \frac{dG}{dt} - \frac{dG}{dt} = 0. \quad (5)$$

Noether’s theorem (the payoff)

$$\frac{dJ_k}{dt} = 0 \quad \implies \quad J_k \text{ is conserved.}$$

What J_k *physically means* depends entirely on *which* symmetry you plugged in. Section 4.3 plugs in the three spacetime symmetries and reads off the three famous conservation laws.

Two easy-to-miss points

- **Two different variations are used.** The master identity (1) comes from a *sloppy* variation (EOM not assumed). The final $dJ_k/dt = 0$ is evaluated *on* a real solution (EOM assumed). Both steps are needed.
- **G is not optional decoration.** For translations/rotations it turns out $G = 0$, but for

time translation $G = L$. Getting G right is what makes J_k truly constant.

2. §4.3 Spacetime Symmetries and Their Noether Currents

Now we just turn the crank of eq. (4) three times. Each row of the table below is one symmetry; the punchline conservation laws are the three pillars of mechanics.

2.1 Spatial translations $\Rightarrow$ linear momentum

Physical statement: there is no special point in space; shifting the origin changes nothing. So the action is invariant under $\vec{q} \rightarrow \vec{q} + \vec{\alpha}$:

$$L[\vec{q} + \vec{\alpha}, \dot{\vec{q}}] = L[\vec{q}, \dot{\vec{q}}]. \quad (6)$$

Three parameters $\{\alpha_1, \alpha_2, \alpha_3\}$ (the shift distances). Because L does not change, $\partial L / \partial \alpha_i|_0 = 0$, so we may take $G = 0$. Also $\partial q'_i / \partial \alpha_k = \delta_{ik}$. Plug into (4):

$$J_k^{\text{sp-trans}} = p_i \delta_{ik} - 0 = p_k \quad \Longrightarrow \quad \frac{d\vec{p}}{dt} = 0. \quad (7)$$

Result 1

Space-translation invariance $\Rightarrow$ conservation of linear momentum $\vec{p}$.

2.2 Rotations $\Rightarrow$ angular momentum

Physical statement: no special orientation in space. Rotations form the group $SO(3)$; for a small rotation (from Ch. 3),

$$q'_i = [e^{i\vec{\alpha} \cdot \vec{J}}]_i^j q_j \simeq q_i - (\vec{\alpha} \times \vec{q})_i, \quad \text{so} \quad \frac{\partial q'_i}{\partial \alpha_k} = \epsilon_{ikj} q_j. \quad (8)$$

A rotationally invariant L can only depend on the *magnitude* $|\vec{q}|$ (rotations preserve lengths), so again $\partial L / \partial \alpha_i|_0 = 0$ and $G = 0$. Then

$$J_k^{\text{rot}} = p_i \epsilon_{ikj} q_j = (\vec{q} \times \vec{p})_k \quad \Longrightarrow \quad \frac{d}{dt}(\vec{q} \times \vec{p}) = 0. \quad (9)$$

Result 2

Rotational invariance $\Rightarrow$ conservation of angular momentum $\vec{q} \times \vec{p}$.

2.3 Time translations $\Rightarrow$ energy

Physical statement: the laws of physics are the same today as yesterday; shifting the clock's zero changes nothing. Under $t' = t + \delta\alpha$,

$$\left. \frac{\partial q'_i}{\partial \alpha} \right|_0 = \dot{q}_i. \quad (10)$$

Just *one* parameter α (how far the time origin moves). This case is different: $G \neq 0$. Because shifting time shifts the whole Lagrangian,

$$\delta_\alpha L = \frac{dL}{dt} \delta\alpha + \dots \Rightarrow \left. \frac{\partial L}{\partial \alpha} \right|_0 = \frac{dL}{dt} \Rightarrow G = L. \quad (11)$$

Plugging $\partial q'_i / \partial \alpha = \dot{q}_i$ and $G = L$ into (4):

$$J^{\text{t-trans}} = p_i \dot{q}_i - L \equiv H \quad \Rightarrow \quad \frac{dH}{dt} = 0. \quad (12)$$

The combination $H = p_i \dot{q}_i - L$ is exactly the **Hamiltonian** (the energy).

Result 3

Time-translation invariance $\Rightarrow$ conservation of energy H .

Why energy conservation means a closed system

Time translation is a symmetry exactly when L has *no explicit* t -dependence. If L depended on t , something external could be pumping the system at a later time, and the action would shift when you move the clock. “No explicit t ” is the definition of a **closed system** — so energy conservation holds for any closed system.

3. One-page recap

The §4.2–4.3 machine

Symmetry	$\frac{\partial q'_i}{\partial \alpha_k}$	G	Conserved J_k
Space translation	δ_{ik}	0	p_k (linear momentum)
Rotation $SO(3)$	$\epsilon_{ikj} q_j$	0	$(\vec{q} \times \vec{p})_k$ (angular momentum)
Time translation	$\dot{q}_i$	L	$H = p_i \dot{q}_i - L$ (energy)

Recipe: (1) write the symmetry and its parameters α ; (2) compute $\partial q' / \partial \alpha$; (3) find G from $\delta_\alpha L = \frac{d}{dt} G$; (4) assemble $J_k = p_i \partial q'_i / \partial \alpha_k - G$; (5) it's automatically conserved on solutions.